Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Now consider the statements I through III.

I. If one of x or y is even, then one of x or y is divisible by 2. It follows that xy is divisible by 2 and that xy is even.

II. Given that a number x or y is odd—not divisible by 2—we know that its product with itself is not divisible by 2 and is therefore odd. On the other hand, given that a number x or y is even, we know that its product with itself *is* divisible by 2 and is therefore even. The sum $x^2 + y^2$ is therefore the sum of an even number and an odd number. In such a case, the sum can be written as $(2m) + (2n + 1) = 2(m + n) + 1$, with m and n integers. It follows that $x^2 + y^2$ is not divisible by 2 and is therefore odd.

III. We know that one of x or y is even and the other is odd. We can therefore see from the discussion of statement II that $x + y$ is odd, and then also see, from the discussion of statement II, that the product of $x + y$ with itself, $(x + y)^2$, is odd.

The correct answer is D.

•

On Monday, the opening price of a certain stock was $100 per share and its closing price was $110 per share. On Tuesday the closing price of the stock was 10 percent less than its closing price on Monday, and on Wednesday the closing price of the stock was 4 percent greater than its closing price on Tuesday. What was the approximate percent change in the price of the stock from its opening price on Monday to its closing price on Wednesday?

A. A decrease of 6%

B. A decrease of 4%

C. A decrease of 1%

D. An increase of 3%

E. An increase of 4%

Arithmetic Percents

The closing share price on Tuesday was 10% less than the closing price on Monday, $110. 10% of $110 is equal to 0.1 × $110 = $11, so the closing price on Tuesday was $110 − $11 = $99. The closing price on Wednesday was 4% greater than this: $99 + (0.04 × $99) = $99 + $3.96 = $102.96. This value, $102.96, is 2.96% greater than $100, the opening price on Monday. The percentage change from the opening share price on Monday is therefore an increase of approximately 3%, which is the closest of the available answers to an increase of 2.96%.

The correct answer is D.

•

1 − 0.000001 =

A. (1.01) (0.99)

B. (1.11) (0.99)

C. (1.001) (0.999)

D. (1.111) (0.999)

E. (1.0101) (0.0909)

Arithmetic Place Value

The task in this question is to find among the available answers the expression that is equal to 1 − 0.000001 = 0.999999. In the case of answer choice C, the first of the two factors, (1.001), is equal to 1 + 0.001. One may therefore observe that (1.001)(0.999) = (1 + 0.001)(0.999) = 0.999 + 0.000999 = 0.999999. Answer choice C is therefore a correct answer.

For answer choice A, (1.01)(0.99) = (1 + 0.01)(0.99) = 0.9999. This answer choice is therefore incorrect. For answer choice B, (1.11)(0.99) = (1 + 0.1 + 0.01)(0.99) = 0.99 + 0.099 + 0.0099 = 1.0989. This answer choice is therefore incorrect. For answer choice D, (1.111)(0.999) = 0.999 + 0.0999 + 0.00999 + 0.000999 = 1.109889. This answer choice is therefore incorrect. For answer choice E, (1.0101)(0.909) = 0.909 + 0.00909 + 0.0000909 = 0.9181809. This answer choice is therefore incorrect.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

answer choice is therefore incorrect.

The correct answer is C.

•

In a certain history class of 17 juniors and seniors, each junior has written 2 book reports and each senior has written 3 book reports. If the 17 students have written a total of 44 book reports, how many juniors are in the class?

A. 7

B. 8

C. 9

D. 10

E. 11

Algebra Simultaneous Equations

Letting j and s, respectively, represent the juniors and seniors in the class, it is given that $j + s = 17$ or $s = 17 - j$. Also, since it is given that each junior has written 2 book reports and each senior has written 3 book reports for a total of 44 book reports, it follows that $2j + 3s = 44$ or $2j + 3(17 - j) = 44$. Therefore, $j = 3(17) - 44 = 7$.

The correct answer is A.

•

$|-4|(|-20| - |5|) =$

A. −100

B. −60

C. 60

D. 75

E. 100

Arithmetic Absolute Value

$|-4|(|-20| - |5|) = 4(20 - 5) = 4 \times 15 = 60$

The correct answer is C.

•

What is the greatest integer k for which 32.45×10^k is less than 1?

A. –2

B. 1

C. 0

D. 1

E. 2

Arithmetic Operations with Integers

$32.45 \times 10^{-2} = 0.3245 < 1$, but $32.45 \times 10^{-1} = 3.245 > 1$. Thus, –2 is the greatest integer k for which $32.45 \times 10^k < 1$.

The correct answer is A.

•

How many integers x satisfy both $2 < x \leq 4$ and $0 \leq x \leq 3$?

A. 5

B. 4

C. 3

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

D. 2

E. 1

Arithmetic Inequalities

The integers that satisfy $2 < x \leq 4$ are 3 and 4. The integers that satisfy $0 \leq x \leq 3$ are 0, 1, 2, and 3. The only integer that satisfies both $2 < x \leq 4$ and $0 \leq x \leq 3$ is 3, and so there is only one integer that satisfies both $2 < x \leq 4$ and $0 \leq x \leq 3$.

The correct answer is E.

•

At the opening of a trading day at a certain stock exchange, the price per share of stock K was \$8. If the price per share of stock K was \$9 at the closing of the day, what was the percent increase in the price per share of stock K for that day?

A. 1.4%

B. 5.9%

C. 11.1%

D. 12.5%

E. 23.6%

Arithmetic Percents

An increase from \$8 to \$9 represents an increase of $\left(\frac{9-8}{8} \times 100\right)\% = \frac{100}{8}\% = 12.5\%$.

The correct answer is D.

•

As shown in the diagram above, a lever resting on a fulcrum has weights of w_1 pounds and w_2 pounds, located d_1 feet and d_2 feet from the fulcrum. The lever is balanced and $w_1 d_1 = w_2 d_2$. Suppose w_1 is 50 pounds and w_2 is 30 pounds. If d_1 is 4 feet less than d_2, what is d_2, in feet?

A. 1.5

B. 2.5

C. 6

D. 10

E. 20

Algebra First-Degree Equations; Substitution

Given $w_1 d_1 = w_2 d_2$, $w_1 = 50$, $w_2 = 30$, and $d_1 = d_2 - 4$, it follows that $50(d_2 - 4) = 30d_2$, and so

$$\begin{aligned} 50(d_2 - 4) &= 30d_2 && \text{given} \\ 50d_2 - 200 &= 30d_2 && \text{distributive principle} \\ 20d_2 &= 200 && \text{add } 200 - 30d_2 \text{ to both sides} \\ d_2 &= 10 && \text{divide both sides by 20} \end{aligned}$$

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

$a_2 = 10$ divide both sides by 20

The correct answer is D.

Day	Change in Dollars
Monday	$+1\frac{1}{2}$
Tuesday	$-\frac{3}{4}$
Wednesday	0
Thursday	$-\frac{1}{8}$
Friday	$+2\frac{1}{4}$

•

The table above shows the daily change in the price of a certain stock last week. What was the net change in dollars in the price of the stock for the week?

A. $-4\frac{5}{8}$

B. $-2\frac{7}{8}$

C. $+2\frac{7}{8}$

D. $+3\frac{3}{4}$

E. $-4\frac{5}{8}$

Arithmetic Interpretation of Tables

The net change in dollars is $\frac{3}{2}-\frac{3}{4}-\frac{1}{8}+\frac{9}{4}=\frac{12-6-1+18}{8}=\frac{23}{8}=2\frac{7}{8}$. So, that's an increase of $2\frac{7}{8}$ dollars.

The correct answer is C.

•

Of 120 hotel rooms rented one night, some were suites rented for $115 each and the rest were double rooms rented for $85 each. If the total revenue from the room rentals for this night was $10,890, how many suites were rented?

A. 23

B. 35

C. 54

D. 94

E. 99

Algebra Simultaneous Equations

Let x be the number of suites rented. Then $\$10,890 = \$115x + \$85(120 - x) = \$115x + \$10,200 - \$85x$, so

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Let x be the number of suites rented. Then $\$10{,}890 = \$115x + \$85(120 - x) = \$115x + \$10{,}200 - \$85x$, so $\$690 = \$30x$. Thus, $x = 23$.

The correct answer is A.

•

Last week Jack worked 70 hours and earned $1,260. If he earned his regular hourly wage for the first 40 hours worked, $1\frac{1}{2}$ times his regular hourly wage for the next 20 hours worked, and 2 times his regular hourly wage for the remaining 10 hours worked, what was his regular hourly wage?

A. $7.00

B. $14.00

C. $18.00

D. $22.00

E. $31.50

Algebra First-Degree Equations

If w represents Jack's regular hourly wage, then Jack's earnings for the week can be represented by the sum of the following amounts, in dollars: $40w$ (his earnings for the first 40 hours he worked), $(20)(1.5w)$ (his earnings for the next 20 hours he worked), and $(10)(2w)$ (his earnings for the last 10 hours he worked). Therefore,

$$\begin{aligned} 40w + (20)(1.5w) + (10)(2w) &= 1{,}260 \quad \text{given} \\ 90w &= 1{,}260 \quad \text{add like terms} \\ w &= 14 \quad \text{divide both sides by 90} \end{aligned}$$

Jack's regular hourly wage was $14.00.

The correct answer is B.

•

If $\frac{x}{4}$ is 2 more than $\frac{x}{8}$, then $x =$

A. 4

B. 8

C. 16

D. 32

E. 64

Algebra First-Degree Equations

If $\frac{x}{4} = 2 + \frac{x}{8}$, then we can multiply both sides of the equation by 8 to find that $2x = 16 + x$, so $x = 16$.

The correct answer is C.

•

If $(2^x)(2^y) = 8$ and $(9^x)(3^y) = 81$, then $(x, y) =$

A. (1, 2)

B. (2, 1)

C. (1, 1)

D. (2, 2)

E. (1, 3)

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. (1, 3)

Algebra Exponents

If $(2^x)(2^y) = 8 = 2^3$, then $x + y = 3$. Of the five answer options, the only ones satisfying this equation are option A, in which $(x, y) = (1, 2)$, and option B, in which $(x, y) = (2, 1)$. Since $(9^x)(3^y) = 81 = (9^1)(3^2) \neq (9^2)(3^1)$, we find that $x = 1$ and $y = 2$, so $(x, y) = (1, 2)$.

The correct answer is A.

•

A certain toll station on a highway has 7 tollbooths, and each tollbooth collects $0.75 from each vehicle that passes it. From 6 o'clock yesterday morning to 12 o'clock midnight, vehicles passed each of the tollbooths at the average rate of 4 vehicles per minute. Approximately how much money did the toll station collect during that time period?

A. $1,500

B. $3,000

C. $11,500

D. $23,000

E. $30,000

Arithmetic Rate Problem

On average, 4 vehicles pass each tollbooth every minute. There are 7 tollbooths at the station, and each passing vehicle pays $0.75. Therefore, the average rate, per minute, at which money is collected by the toll station is $\$(7 \times 4 \times 0.75) = \$(7 \times 4 \times \frac{3}{4}) = \$(7 \times 3) = \$21$. From 6:00 a.m. through midnight there are 18 hours. And because 18 hours is equal to 18 × 60 minutes, from 6:00 a.m. through midnight there are 1,080 minutes. The total amount of money collected by the toll station during this period is therefore 1,080 × $21 = $22,680, which is approximately $23,000.

The correct answer is D.

•

If $y = 2 + 2k$ and $y \neq 0$, then $\frac{1}{y} + \frac{1}{y} + \frac{1}{y} + \frac{1}{y} =$

A. $\frac{1}{8 + 8k}$

B. $\frac{2}{1 + k}$

C. $\frac{1}{8 + k}$

D. $\frac{4}{8 + k}$

E. $\frac{4}{1 + k}$

Algebra Equations

If $y = 2 + 2k \neq 0$, then $\frac{1}{y} + \frac{1}{y} + \frac{1}{y} + \frac{1}{y} = \frac{4}{y} = \frac{4}{2 + 2k} = \frac{2}{1 + k}$.

The correct answer is B.

•

If $3^x - 3^{x-1} = 162$, then $x(x - 1) =$

A. 12

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

A. 12

B. 16

C. 20

D. 30

E. 81

Algebra Exponents

If $3^x - 3^{x-1} = 162 = 2(81) = 2(3^4) = 3(3^4) - 1(3^4) = 3^5 - 3^4$, then $x = 5$. Thus, $x(x-1) = 5(4) = 20$.

The correct answer is C.

•

The expression $n!$ is defined as the product of the integers from 1 through n. If p is the product of the integers from 100 through 299 and q is the product of the integers from 200 through 299, which of the following is equal to $\frac{p}{q}$?

A. 99!

B. 199!

C. $\frac{199!}{99!}$

D. $\frac{299!}{99!}$

E. $\frac{299!}{199!}$

Arithmetic Series and Sequences

The number p is equal to 100 × 101 × 102 × … × 299 and the number q is equal to 200 × 201 × 202 × … × 299. The number $\frac{p}{q}$ is thus equal to

$$\frac{100 \times 101 \times 102 \times \ldots \times 299}{200 \times 201 \times 202 \times \ldots \times 299} = \frac{100 \times 101 \times 102 \times \ldots \times 199 \times 200 \times 201 \times 202 \times \ldots \times 299}{200 \times 201 \times 202 \times \ldots \times 299}.$$

Canceling 200 × 201 × 202 × … × 299 from the numerator and the denominator, we see that $\frac{p}{q} = 100 \times 101 \times 102 \times \ldots \times 199$. Note that the multiplication in this expression for $\frac{p}{q}$ begins with 100 (the smallest of the numbers being multiplied), whereas the multiplication in $n! = 1 \times 2 \times 3 \times \ldots \times n$ begins with 1. Starting with 199! as our numerator, we thus need to find a denominator that will cancel the undesired elements of the multiplication (in 199!). This number is 1 × 2 × 3 × … × 99 = 99! That is,

$$\frac{p}{q} = 100 \times 101 \times 102 \times \ldots \times 199 = \frac{1 \times 2 \times 3 \times \ldots \times 99 \times 100 \times 101 \times 102 \times \ldots \times 199}{1 \times 2 \times 3 \times \ldots \times 99} = \frac{199!}{99!}.$$

The correct answer is C.

•

A school club plans to package and sell dried fruit to raise money. The club purchased 12 containers of dried fruit, each containing $16\frac{3}{4}$ pounds. What is the maximum number of individual bags of dried fruit, each containing $\frac{1}{4}$ pounds, that can be sold from the dried fruit the club purchased?

A. 50

B. 64

C. 67

D. 768

E. 804

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. 804

Arithmetic Applied Problems; Operations with Fractions

The 12 containers, each containing $16\frac{3}{4}$ pounds of dried fruit, contain a total of

$(12)\left(16\frac{3}{4}\right) = (12)\left(\frac{67}{4}\right) = (3)(67) = 201$ pounds of dried fruit, which will make $\frac{201}{\frac{1}{4}} = (201)(4) = 804$ individual bags that can be sold.

The correct answer is E.

Height	Price
Less than 5 ft	$14.95
5 ft to 6 ft	$17.95
Over 6 ft	$21.95

•

A nursery sells fruit trees priced as shown in the chart above. In its inventory 54 trees are less than 5 feet in height. If the expected revenue from the sale of its entire stock is estimated at $2,450, approximately how much of this will come from the sale of trees that are at least 5 feet tall?

A. $1,730

B. $1,640

C. $1,410

D. $1,080

E. $810

Arithmetic Applied Problems

If the nursery sells its entire stock of trees, it will sell the 54 trees that are less than 5 feet in height at the price per tree of $14.95 shown in the chart. The expected revenue from the sale of the trees that are less than 5 feet tall is therefore 54 × $14.95 = $807.30. The revenue from the sale of the trees that are at least 5 feet tall is thus equal to the total revenue from the sale of the entire stock of trees minus $807.30. The revenue from the sale of the entire stock of trees is estimated at $2,450. Based on this estimate, the revenue from the sale of the trees that are at least 5 feet tall will be $2,450 − $807.30 = $1,642.70, which is approximately $1,640.

The correct answer is B.

•

A certain bridge is 4,024 feet long. Approximately how many minutes does it take to cross this bridge at a constant speed of 20 miles per hour? (1 mile = 5,280 feet)

A. 1

B. 2

C. 4

D. 6

E. 7

Arithmetic Applied Problems

First, convert 4,024 feet to miles since the speed is given in miles per hour: $4{,}024\ \cancel{ft} \times \frac{1\ \text{mi}}{5{,}280\ \cancel{ft}} = \frac{4{,}024}{5{,}280}\ \text{mi}.$

$\frac{4{,}024}{\ }\ \frac{20\ \text{mi}}{\ }\ \frac{4{,}024\ \cancel{\text{mi}}}{\ }\ \frac{1\ \text{hr}}{\ }\ \frac{4{,}024\ \text{hr}}{\ }$

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Now, divide by 20 mph: $\frac{4{,}024}{5{,}280}\text{ mi} \div \frac{20\text{ mi}}{1\text{ hr}} = \frac{4{,}024\ \cancel{\text{mi}}}{5{,}280} \times \frac{1\text{ hr}}{20\ \cancel{\text{mi}}} = \frac{4{,}024\text{ hr}}{(5{,}280)(20)}$.

Last, convert $\frac{4{,}024\text{ hr}}{(5{,}280)(20)}$ to minutes:

$\frac{4{,}024\ \cancel{\text{hr}}}{(5{,}280)(20)} \times \frac{60\text{ min}}{1\ \cancel{\text{hr}}} = \frac{(4{,}024)(60)\text{ min}}{(5{,}280)(20)} \approx \frac{4{,}000}{5{,}000} \times \frac{60}{20}\text{ min}$. Then, $\frac{4{,}000}{5{,}000} \times \frac{60}{20}\text{ min} = 0.8 \times 3\text{ min} \approx 2\text{ min}$. Thus, at a constant speed of 20 miles per hour, it takes approximately 2 minutes to cross the bridge.

The correct answer is B.

•

A purse contains 57 coins, all of which are nickels, dimes, or quarters. If the purse contains x dimes and 8 more nickels than dimes, which of the following gives the number of quarters the purse contains in terms of x?

A. $2x - 49$

B. $2x + 49$

C. $2x - 65$

D. $49 - 2x$

E. $65 - 2x$

Algebra First-Degree Equations

Letting Q be the number of quarters, there are $(x + 8)$ nickels, x dimes, and Q quarters for a total of 57 coins.

$$(x + 8) + x + Q = 57 \quad \text{given}$$

$$2x + 8 + Q = 57 \quad \text{combine like terms}$$

$$Q = 49 - 2x \quad \text{subtract } 2x + 8 \text{ from both sides}$$

The correct answer is D.

•

The annual interest rate earned by an investment increased by 10 percent from last year to this year. If the annual interest rate earned by the investment this year was 11 percent, what was the annual interest rate last year?

A. 1%

B. 1.1%

C. 9.1%

D. 10%

E. 10.8%

Arithmetic Percents

If L is the annual interest rate last year, then the annual interest rate this year is 10% greater than L, or $1.1L$. It is given that $1.1L = 11\%$. Therefore, $L = \frac{11\%}{1.1} = 10\%$. (Note that if the given information had been that the investment increased by *10 percentage points*, then the equation would have been $L + 10\% = 11\%$.)

The correct answer is D.

•

A total of 5 liters of gasoline is to be poured into two empty containers with capacities of 2 liters and 6 liters, respectively, such that both containers will be filled to the same percent of their respective capacities. What amount of gasoline, in liters, must be poured into the 6-liter container?

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

A. $4\frac{1}{2}$

B. 4

C. $3\frac{3}{4}$

D. 3

E. $1\frac{1}{4}$

Algebra Ratio and Proportion

If x represents the amount, in liters, of gasoline poured into the 6-liter container, then $5 - x$ represents the amount, in liters, of gasoline poured into the 2-liter container. After the gasoline is poured into the containers, the 6-liter container will be filled to $\left(\frac{x}{6} \times 100\right)\%$ of its capacity and the 2-liter container will be filled to $\left(\frac{5-x}{2} \times 100\right)\%$ of its capacity. Because these two percents are equal,

$$\frac{x}{6} = \frac{5-x}{2} \quad \text{given}$$

$$2x = 6(5-x) \quad \text{multiply both sides by 12}$$

$$2x = 30 - 6x \quad \text{use distributive property}$$

$$8x = 30 \quad \text{add } 6x \text{ to both sides}$$

$$x = 3\frac{3}{4} \quad \text{divide both sides by 8}$$

Therefore, $3\frac{3}{4}$ liters of gasoline must be poured into the 6-liter container.

The correct answer is C.

•

What is the larger of the 2 solutions of the equation $x^2 - 4x = 96$?

A. 8

B. 12

C. 16

D. 32

E. 100

Algebra Second-Degree Equations

It is given that $x^2 - 4x = 96$, or $x^2 - 4x - 96 = 0$, or $(x-12)(x+8) = 0$. Therefore, $x = 12$ or $x = -8$, and the larger of these two numbers is 12.

Alternatively, from $x^2 - 4x = 96$ it follows that $x(x-4) = 96$. By inspection, the left side is either the product of 12 and 8, where the value of x is 12, or the product of −8 and −12, where the value of x is −8, and the larger of these two values of x is 12.

The correct answer is B.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

$$x = \frac{1}{6}gt^2$$

•

In the formula shown, if g is a constant and $x = -6$ when $t = 2$, what is the value of x when $t = 4$?

A. –24

B. –20

C. –15

D. 20

E. 24

Algebra Formulas

Since $x = -6$ when $t = 2$, it follows that $-6 = \left(\frac{1}{6}\right)(g)(4)$ so $g = \frac{3}{2}(-6) = -9$. Then, when $t = 4$, $x = \frac{1}{6}(-9)(16) = -24$.

The correct answer is A.

•

$\frac{(39{,}897)(0.0096)}{198.76}$ is approximately

A. 0.02

B. 0.2

C. 2

D. 20

E. 200

Arithmetic Estimation

$$\frac{(39{,}897)(0.0096)}{198.76} \approx \frac{(40{,}000)(0.01)}{200} = (200)(0.01) = 2$$

The correct answer is C.

•

The "prime sum" of an integer n greater than 1 is the sum of all the prime factors of n, including repetitions. For example, the prime sum of 12 is 7, since 12 = 2 × 2 × 3 and 2 + 2 + 3 = 7. For which of the following integers is the prime sum greater than 35?

A. 440

B. 512

C. 620

D. 700

E. 750

Arithmetic Properties of Numbers

A. Since 440 = 2 × 2 × 2 × 5 × 11, the prime sum of 440 is 2 + 2 + 2 + 5 + 11 = 22, which is not greater than 35.

B. Since 512 = 2^9, the prime sum of 512 is 9(2) = 18, which is not greater than 35.

C. Since 620 = 2 × 2 × 5 × 31, the prime sum of 620 is 2 + 2 + 5 + 31 = 40, which is greater than 35.

Because there can be only one correct answer, D and E need not be checked. However, for completeness,

D. Since 700 = 2 × 2 × 5 × 5 × 7, the prime sum of 700 is 2 + 2 + 5 + 5 + 7 = 21, which is not greater than 35.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. Since 750 = 2 × 3 × 5 × 5 × 5, the prime sum of 750 is 2 + 3 + 5 + 5 + 5 = 20, which is not greater than 35.

The correct answer is C.

•

Each machine at a toy factory assembles a certain kind of toy at a constant rate of one toy every 3 minutes. If 40 percent of the machines at the factory are to be replaced by new machines that assemble this kind of toy at a constant rate of one toy every 2 minutes, what will be the percent increase in the number of toys assembled in one hour by all the machines at the factory, working at their constant rates?

A. 20%

B. 25%

C. 30%

D. 40%

E. 50%

Arithmetic Applied Problems; Percents

Let n be the total number of machines working. Currently, it takes each machine 3 minutes to assemble 1 toy, so each machine assembles 20 toys in 1 hour and the total number of toys assembled in 1 hour by all the current machines is $20n$. It takes each new machine 2 minutes to assemble 1 toy, so each new machine assembles 30 toys in 1 hour. If 60% of the machines assemble 20 toys each hour and 40% assemble 30 toys each hour, then the total number of toys produced by the machines each hour is $(0.60n)(20) + (0.40n)(30) = 24n$. The percent increase in hourly production is $\frac{24n - 20n}{20n} = \frac{1}{5}$ or 20%.

The correct answer is A.

•

When a subscription to a new magazine was purchased for m months, the publisher offered a discount of 75 percent off the regular monthly price of the magazine. If the total value of the discount was equivalent to buying the magazine at its regular monthly price for 27 months, what was the value of m?

A. 18

B. 24

C. 30

D. 36

E. 48

Algebra Percents

Let P represent the regular monthly price of the magazine. The discounted monthly price is then $0.75P$. Paying this price for m months is equivalent to paying the regular price for 27 months. Therefore, $0.75mP = 27P$, and so $0.75m = 27$. It follows that $m = \frac{27}{0.75} = 36$.

The correct answer is D.

•

At a garage sale, all of the prices of the items sold were different. If the price of a radio sold at the garage sale was both the 15th highest price and the 20th lowest price among the prices of the items sold, how many items were sold at the garage sale?

A. 33

B. 34

C. 35

D. 36

E. 37

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. 37

Arithmetic Operations with Integers

If the price of the radio was the 15th highest price, there were 14 items that sold for prices higher than the price of the radio. If the price of the radio was the 20th lowest price, there were 19 items that sold for prices lower than the price of the radio. Therefore, the total number of items sold is 14 + 1 + 19 = 34.

The correct answer is B.

•

Half of a large pizza is cut into 4 equal-sized pieces, and the other half is cut into 6 equal-sized pieces. If a person were to eat 1 of the larger pieces and 2 of the smaller pieces, what fraction of the pizza would remain uneaten?

A. $\frac{5}{12}$

B. $\frac{13}{24}$

C. $\frac{7}{12}$

D. $\frac{2}{3}$

E. $\frac{17}{24}$

Arithmetic Operations with Fractions

Each of the 4 equal-sized pieces represents $\frac{1}{8}$ of the whole pizza since each slice is $\frac{1}{4}$ of $\frac{1}{2}$ of the pizza. Each of the 6 equal-sized pieces represents $\frac{1}{12}$ of the whole pizza since each slice is $\frac{1}{6}$ of $\frac{1}{2}$ of the pizza. The fraction of the pizza remaining after a person eats one of the larger pieces and 2 of the smaller pieces is

$$1-\left[\frac{1}{8}+2\left(\frac{1}{12}\right)\right]=1-\left(\frac{1}{8}+\frac{1}{6}\right)=1-\frac{6+8}{48}=1-\frac{7}{24}=\frac{17}{24}.$$

The correct answer is E.

•

If $a=1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}$ and $b=1+\frac{1}{4}a$, then what is the value of $a-b$?

A. $-\frac{85}{256}$

B. $-\frac{1}{256}$

C. $-\frac{1}{4}$

D. $\frac{125}{256}$

E. $\frac{169}{256}$

Arithmetic Operations with Fractions

Given that $a=1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}$, it follows that $\frac{1}{4}a=\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\frac{1}{256}$ and so $b=1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\frac{1}{256}$. Then

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

$$a-b=\left(1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\frac{1}{256}\right)-\left(1+\frac{1}{4}+\frac{1}{16}+\frac{1}{64}+\frac{1}{256}\right)=-\frac{1}{256}.$$

The correct answer is B.

•

In a certain learning experiment, each participant had three trials and was assigned, for each trial, a score of either –2, –1, 0, 1, or 2. The participant's final score consisted of the sum of the first trial score, 2 times the second trial score, and 3 times the third trial score. If Anne received scores of 1 and –1 for her first two trials, not necessarily in that order, which of the following could NOT be her final score?

A. –4

B. –2

C. 1

D. 5

E. 6

Arithmetic Applied Problems

If x represents Anne's score on the third trial, then Anne's final score is either $1+2(-1)+3x=3x-1$ or $-1+2(1)+3x=3x+1$, where x can have the value –2, –1, 0, 1, or 2. The following table shows Anne's final score for each possible value of x.

x	$3x-1$	$3x+1$
–2	–7	–5
–1	–4	–2
0	–1	1
1	2	4
2	5	7

Among the answer choices, the only one not found in the table is 6.

The correct answer is E.

•

For all positive integers m and v, the expression $m \Theta v$ represents the remainder when m is divided by v. What is the value of ((98 Θ 33) Θ 17) − (98 Θ (33 Θ 17))?

A. −10

B. −2

C. 8

D. 13

E. 17

Arithmetic Operations with Integers

First, for ((98 Θ 33) Θ 17), determine 98 Θ 33, which equals 32, since 32 is the remainder when 98 is divided by 33 (98 = 2(33) + 32). Then, determine 32 Θ 17, which equals 15, since 15 is the remainder when 32 is divided by 17 (32 = 1(17) + 15). Thus, ((98 Θ 33) Θ 17) = 15.

Next, for (98 Θ (33 Θ 17)), determine 33 Θ 17, which equals 16, since 16 is the remainder when 33 is divided by 17 (33 = 1(17) + 16). Then, determine 98 Θ 16, which equals 2, since 2 is the remainder when 98 is divided by 16 (98 =

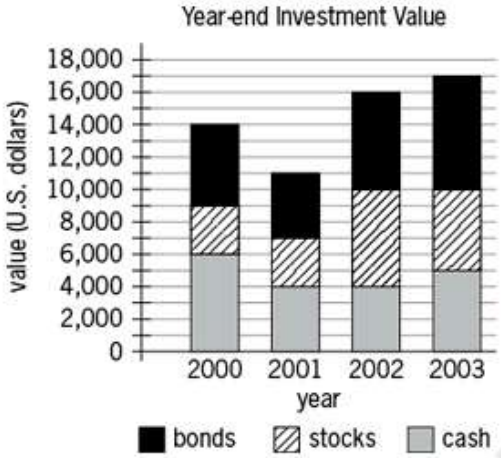
DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

(33 = 1(17) + 16). Then, determine 98 Θ 16, which equals 2, since 2 is the remainder when 98 is divided by 16 (98 = 6(16) + 2). Thus, (98 Θ (33 Θ 17)) = 2.

Finally, ((98 Θ 33) Θ 17 − (98 Θ (33 Θ 17)) = 15 − 2 = 13.

The correct answer is D.



•

The chart above shows year-end values for Darnella's investments. For just the stocks, what was the increase in value from year-end 2000 to year-end 2003?

A. $1,000

B. $2,000

C. $3,000

D. $4,000

E. $5,000

Arithmetic Interpretation of Graphs

From the graph, the year-end 2000 value for stocks is 9,000 − 6,000 = 3,000 and the year-end 2003 value for stocks is 10,000 − 5,000 = 5,000. Therefore, for just the stocks, the increase in value from year-end 2000 to year-end 2003 is 5,000 − 3,000 = 2,000.

The correct answer is B.

•

If the sum of the reciprocals of two consecutive odd integers is $\frac{12}{35}$, then the greater of the two integers is

A. 3

B. 5

C. 7

D. 9

E. 11

Arithmetic Operations with Fractions

The sum of the reciprocals of 2 integers, a and b, is $\frac{1}{a}+\frac{1}{b}=\frac{a+b}{ab}$. Therefore, since $\frac{12}{35}$ is the sum of the reciprocals of 2 consecutive odd integers, the integers must be such that their sum is a multiple of 12 and their product is the same multiple of 35 so that the fraction reduces to $\frac{12}{35}$. Considering

DO NOT COPY filippo.fiz@uptoten.it